much more apt to command loyalty if they were easy. A possible interpretation of these findings is that the most natural and consistent approach for a self-confident person is a firm one, whereas more lenient supervision is more natural for an individual lacking self-confidence. Some supervisors may feel constrained to use an approach to their subordinates that is not suited to their personalities—perhaps because of pressures from superiors, perhaps as a result of having acquired certain conceptions as to the "proper" conduct for a supervisor, such as that a supervisor should be easy-going and foster "good human relations" with subordinates, or, conversely, that he should be firm and should maintain social distance. It is probable that a person who self-consciously adopts a style of supervision that is at variance with his normal mode of conduct will feel uncomfortable and behave erratically on occasion, whereas one whose approach to subordinates does not conflict with his personality and usual demeanor will be more at ease and consistent.[48] Supervisors whose approach is inconsistent with their dispositions, whether they are strict or lenient, are apparently less likely to command the loyalty of subordinates than consistent supervisors. This conclusion may also account for the earlier-mentioned finding that supervisory detachment is associated with work-group loyalty. A detached supervisor is undoubtedly more consistent in his behavior toward subordinates than an excitable one, and this consistency may well be a main reason why he is more apt to win their loyalty.

Hierarchical and Peer Relations

Testing a Hypothesis. Much of the work done in the field of formal organizations emphasizes two different facets of these social systems: the impersonality and emotional detachment of relations between individuals and groups on different hierarchical levels, and the cohesiveness and informality of relations among peers on the same organizational level. This difference suggests a possible causal link between peer-group cohesion and hierarchical social distance. The following hypothesis may thus be offered: an official who enjoys social integration among his peers will be more impersonal and detached in his relations with subordinates than one who does not. For peer-group support allows the superior to be more detached by obviating the need for courting the support of his subordinates. To test this hypothesis, we used data from both welfare agencies. Specifically, we examined the association between the index of detachment to subordinates already

[48] In this connection, note Likert's comment: "A subordinate tends also to expect his superior to behave in ways consistent with the personality of the superior." Likert, *op. cit.*, p. 191.

described and three measures of social integration among peers: whether the supervisor felt integrated among the other supervisors in his department; the number of "friendship" choices he received from other supervisors in his department; and the total number of such choices he received from all supervisors in the agency.

The supervisor's social integration among other supervisors, regardless of which of the three measures was used, was not related to detachment toward subordinates in either agency. The hypothesis that peer-group integration makes supervisors more detached in their relations with caseworkers is clearly negated. Data from City Agency allowed us to determine whether or not peer-group integration among supervisors promoted independence from superiors. Again, the hypothesis was not supported by the data. Integrated supervisors were no more likely than others to exhibit hierarchical independence.

How can we account for this rejection of the hypothesis? With the benefit of hindsight, we realize now that an implicit assumption underlies our hypothesis, and this assumption proved to be incorrect. We had assumed that integrative peer relations and group solidarity tend to develop among supervisors, just as they often do among workers. But this tendency evidently was not present among the supervisors in the two organizations studied. Although supervisors had much more seniority on the average than workers, and seniority was related to integration among workers, supervisors tended to be much less integrated in networks of social relations with peers than did workers. Only three of the twelve supervisors in City Agency received two or more friendship choices from other supervisors, but exactly half of the 60 workers in this agency received two or more choices from colleagues; similarly, in County Agency, only one of the eleven supervisors received as many as two friendship choices, while more than half of the 87 workers did. In City Agency over one-third of the workers, but none of the supervisors, reported that they saw other members of the agency staff socially after working hours as often as once a month. Similar disparities were found for County Agency, where almost two-fifths of the workers but only one of the eleven supervisors reported that they saw other members of the agency socially as frequently as once a month. Social relations among supervisors in the same section were particularly weak, and this was especially pronounced for consultation. In City Agency, six of the ten supervisors interviewed said that they never consulted a colleague (supervisor) in their own section, whereas only 12 out of 60 workers said that they never consulted another worker in their own work group. Supervisors in County Agency were not asked about their consultation practices, but the observation record on the frequency of social interaction provides some clues about the extent of consultation

among supervisors compared with that among caseworkers. On the average, caseworkers initiated 3.8 contacts with other workers in their own division per hour, whereas supervisors initiated only 0.9 hourly contacts with other supervisors. The difference in City Agency was even more pronounced, initiated ingroup contacts per hour being 4.8 for caseworkers and 0.3 for supervisors.[49]

In short, our findings indicate that supervisors in the two agencies tended to be somewhat isolated from supportive contacts with peers. Jaques, to name only one among several observers, has called attention to the isolated position of the top managers in large organizations;[50] but it is rather surprising to find comparable isolation for these first-line supervisors. The supervisors in a given section worked under the same superior, had weekly meetings as a group, and spent all day in close proximity in the office. Nevertheless, solidary bonds uniting them into a group hardly developed among them. Of special significance was the finding that they rarely consulted one another. Perhaps their conception of the supervisory role as being responsible for advising others discouraged them from seeking advice themselves. Perhaps, too, being a link in a hierarchical chain focused the supervisor's attention upon his superiors and subordinates rather than his peers. This suggestion receives some confirmation from the finding that none of the ten supervisors interviewed in City Agency felt that it was more important to be highly thought of by his peers than by his superior or his subordinates, whereas nearly one-third of the workers (19 of 60) considered the good opinion of their peers more important than those of either their supervisor or their clients.[51] Among these supervisors, then, there was no concern with winning the respect of their peers. And virtually no cooperative relations, such as consultation, developed among them. As a result, social relations among colleagues did not furnish sufficient social support for supervisors to have any bearing on their detachment from subordinates or their independence from superiors. This isolation was in contrast to the more significant peer relations that developed in work groups on the bottom level, which seemed to further detachment and independence from clients, as was discussed in Chapters III and IV.

[49] County Agency data are based on the observation of 54 workers and 7 superiors for 28 hours, and City Agency data are based on the observation of 22 workers and 4 supervisors for 24 hours.

[50] Elliott Jaques, *The Changing Culture of a Factory,* New York: Dryden, 1952, pp. 278–279.

[51] Although the difference between workers and supervisors is not as great, these findings are supported by roughly comparable data from County Agency. Workers and supervisors were asked from which of several sources they obtained the most intellectual and professional stimulation. Some workers (8 of 86) but none of the (11) supervisors chose their peer group on this criterion.

Loyalty of Subordinates and Loyalty to the Superior. What, then, is the source of social support that enables some supervisors to maintain detachment and independence? It may well be the loyalty of subordinates.[52] Detached supervisors had more loyal subordinates, as was previously shown. This relation is undoubtedly in part due to the adverse effects of excitability, and of the inconsistency it entails, on loyalty. But it may also result in part from the fact that commanding the allegiance and respect of subordinates is a source of strength for the supervisor that facilitates calm detachment. The supervisor's hierarchical independence, too, was found to be associated with high loyalty of his subordinates. The qualities of the supervisor that allow him to remain independent from his superior probably command the respect and loyalty of his subordinates, as we have suggested. But it may also be that the loyalty of subordinates helps the supervisor to attain and maintain his independence from his superior. This interpretation receives some support from the interview responses of ten City Agency supervisors.

Supervisors who commanded the loyalty of their subordinates were less likely than those who did not to express loyalty to their own superior (the section chief). None of the four supervisors whose subordinates did express high loyalty to them felt loyal to his own superior, while five of the six supervisors who did not command high loyalty from their subordinates expressed loyalty to their section chief. And although detachment from subordinates and hierarchical independence were not related to one another,[53] both were inversely related to the supervisor's loyalty to his superior.

The loyalty of subordinates thus appears to have been a source of social support for first-line supervisors. Social support from subordinates probably lessens the supervisor's need to seek the support and approval of his superior by becoming attached to him and emulating his style of supervision. Conversely, strong ties of loyalty to one's superior may reduce the need of the supervisor to win the respect and allegiance of his own subordinates.[54] This latter interpretation implies that loyalty to superiors in a hierarchical organization would be pronounced on alternate levels. If a manager does not command the loyalty of the section chiefs under him, it will be particularly important for them to obtain social support by commanding the loyalty of their subordinates, the first-line supervisors. And if a section chief is successful

[52] This discussion is based entirely on data from City Agency since no loyalty measures were available in the County study.

[53] Four out of five independent supervisors were detached toward their subordinates, but so were three out of seven dependent supervisors.

[54] Here again our survey data require us to present alternative interpretations of observed relations inasmuch as they do not allow us to determine the direction of influence between the variables.

in establishing supportive working relationships with the supervisors under him, the incentive of these supervisors to win the allegiance and respect of their subordinates will be reduced.[55] Hence, the orientations of alternate hierarchical levels would be similar, and those of adjacent levels, different. Although this conclusion is highly speculative, it parallels Caudill's observation that alternate levels in the hierarchy of the hospital he studied were more similar in orientation than adjacent ones.[56]

Concluding Remarks

Different styles of supervision apparently exert a variety of influences on workers. Emotional detachment, consistency, and hierarchical independence were found to be the supervisory characteristics most closely related to the ability to command the loyalty of subordinates. Authoritarian practices on the part of the supervisor, on the other hand, had no bearing on loyalty in the welfare agencies. Worker productivity was largely associated with factors that were related to loyalty, perhaps because supervisors who had won the loyal support of their subordinates were most successful in commanding willing compliance with their directives and in stimulating effort in their work group. Authoritarian practices by the supervisor adversely affected work satisfaction, the willingness to assume responsibility, and the tendency to extend casework service to clients. But in contrast to the findings obtained in other types of organizations, authoritarian supervision did not lower productivity in the two welfare agencies. The supervisory attributes that exerted most influence on group solidarity were hierarchical independence and consistency.

Contrary to expectations, the supervisor's relations with his peers did not affect his approach to subordinates, probably because only a low level of peer-group solidarity existed among supervisors. Workers seemed typically to derive their social support from cooperative relations with their peers, notably from consulting one another. Supervisors, on the other hand, obtained their social support not from working relations with other supervisors, but from loyal subordinates or from their relations with their superiors. The data suggest that a supervisor who expresses loyalty to his superior is less apt than another to command the loyalty of his own subordinates, to be detached in his relations with subordinates, and to exhibit hierarchical inde-

[55] But this statement would not hold true for caseworkers organized in work groups, since they have not two but three alternative sources of social support in the work situation—their peers, their superiors, and their clients.

[56] Caudill, *op. cit.*, pp. 155–157.

pendence. If this suggestion is generally true, it would imply that a manager's ability to command his subordinates' loyalty impedes their effectiveness as supervisors in their own right. This implication, in turn, suggests that effective management can be expected to occur only on alternate levels in the hierarchy.

CHAPTER SEVEN

Managerial Control

The discussions of management that appear in the literature differ considerably, because they tend to emphasize different facets of this complex topic. Let us briefly examine some of the implicit models of management underlying these discussions, without attempting to provide an exhaustive list or assuming that these conceptions are mutually exclusive.

One model views management as "executive leadership" and focuses on such functions as defining objectives, setting policies, and inspiring effort. The writings of Barnard exemplify this emphasis; for him the essence of executive responsibility is the ability "to bind the wills of men to the accomplishment of purposes beyond their immediate ends, beyond their times."[1] Selznick, too, views leadership as the key function of management and suggests that it is required for making critical as opposed to routine decisions—decisions that define the ends of the organization and constitute commitments that shape its essential character.[2] He considers administrative leaders responsible not only for making such "character-defining" decisions but also for seeing that the organization embodies and implements these purposes once they have been established, for defending the integrity of the institution against internal changes and external attacks, and for maintaining order and control over internal conflicts.

A second conception of management emphasizes the hierarchy of "bureaucratic authority" in the organization. The focus here is on impersonal discipline and rational expertness, and on control through directives originated at the apex of the authority pyramid and transmitted through channels down to its base. This, of course, is Weber's model of management. As has been suggested, there is a possible conflict between compliance with directives and the exercise of professional judgment, and the implications of this conflict have received increasing attention from recent students of administrative structures, such as Parsons and Gouldner.

1 Chester I. Barnard, *The Functions of the Executive,* Cambridge, Mass.: Harvard University Press, 1938, p. 283.

2 Philip Selznick, *Leadership in Administration,* Evanston, Ill.: Row, Peterson, 1957, pp. 30–37, 62–64.

A third model conceives of management as the "regulator of the incentive systems." The basic assumptions of this approach are that contributions are made to the organization for rewards received and that management induces contributions by regulating the structure of rewards. Attention was at first directed almost entirely to financial incentives; however, empirical studies as to their effects have been inconclusive. For example, a study by Dale of 305 individual wage-incentive arrangements in 29 different industries reports that wage incentives were found to improve productivity,[3] whereas a study by Behrend of 250 British firms found no evidence in support of the effectiveness of piece-work incentive payments.[4] And Roberts found that the compensation of the highest-paid official in a company was related to the company's size but not to its profit.[5] More recent conceptions of incentives have emphasized the significance of other rewards as well as monetary ones. Thus Whyte and his associates reject the simple model of economic man and stress that incentive systems must be broadly defined as including worker-company relations, intergroup work relations, interaction patterns, and symbols.[6] Likert has attempted to specify the interrelations between several factors that have been demonstrated to influence worker motivation.[7] In addition to distinguishing between monetary and nonmonetary rewards, some writers have distinguished between incentive systems that reward results and those that reward conformity with officially established procedures. Worthy has suggested that the former system "develops initiative and self-reliance and generates a far more powerful driving force than could ever be imposed from the top down."[8] Dimock also concludes that the rewarding of results promotes initiative as well as responsibility in performance.[9]

A fourth model conceptualizes management as "centralized plan-

[3] John D. Dale, *Wage Incentives and Productivity,* New York: George Elliott, 1958.

[4] Hilde Behrend, "Financial Incentives as the Expression of a System of Beliefs," *The British Journal of Sociology,* 10 (1959), pp. 137–147.

[5] David R. Roberts, "A General Theory of Executive Compensation Based on Statistically Tested Propositions," *Quarterly Journal of Economics,* 70 (1956), pp. 270–294. For further discussion of these findings, see Herbert A. Simon, "The Compensation of Executives," *Sociometry,* 20 (1957), pp. 32–35. And for an analysis of the general problem, see David R. Roberts, *Executive Compensation,* Glencoe, Ill.: Free Press, 1959.

[6] William F. Whyte *et al., Money and Motivation,* New York: Harper, 1955.

[7] Rensis Likert, "A Motivational Approach to a Modified Theory of Organization and Management," Mason Haire (ed.), *Modern Organization Theory,* New York: Wiley, 1959, pp. 184–217.

[8] James C. Worthy, "Organizational Structure and Employee Morale," *American Sociological Review,* 15 (1950), p. 178.

[9] Marshall E. Dimock, *Administrative Vitality,* New York: Harper, 1959, pp. 189–191.

ning"—the coordination of operations through advance planning of the work program. Some aspects of this focus are included in the foregoing models: the leader makes critical decisions relating to the over-all design of the enterprise, coordination of activities is a basic function of the bureaucratic hierarchy, and incentive systems are designed to coordinate efforts in the interest of effective achievement of organizational purposes. But the specific emphasis of the planning model is on creating conditions in the organization that influence its members rather than on influencing them directly. This distinction is illustrated by differences between job-lot and assembly-line production as discussed by Richardson and Walker.[10] In the former, which is the traditional organization of production, the coordination of specialized tasks of various sections depends on directives issued by management and transmitted through the hierarchy. But for the assembly line, the flow of production has been organized by management in advance to serve as the coordinating mechanism that obviates the need for much direct supervision.

This brief overview of various models underlying some of the treatments of management will serve as a background for our discussion of this topic. In this chapter we shall first consider the shape of the hierarchical pyramid and then examine social relations among managers. In the second part we shall analyze impersonal mechanisms of control and their significance for the management of formal organizations. Finally, we shall question some of the prevailing assumptions in the study of management.

The Hierarchy

The Shape of the Pyramid. While the significance of social relations for supervision has been particularly emphasized in more recent research, it was by no means entirely ignored by earlier students of administration. Graicunas, for example, called attention to some important implications for management of the fact that as the number of persons in a group increases the number of possible relations between group members increases at a much faster rate.[11] Thus, in a group of 2 persons there is only 1 social relation, while in a group of 6 there are 15 relations between members. Graicunas felt that inasmuch as the manager's responsibility requires an understanding of

10 Frederick L. Richardson and Charles R. Walker, *Human Relations in an Expanding Company,* New Haven: Labor and Management Center, Yale University, 1948, pp. 54–85.

11 The rate is $N\,(N-1)/2$, when N is the number of group members. V. A. Graicunas, "Relationship in Organization," Luther Gulick and L. Urwick (eds.), *Papers on the Science of Administration,* New York: Institute of Public Administration, 1937, pp. 183–187.

the social relations among subordinates as well as of his own with them, no manager should be in charge of more than a small number of subordinates. Many subsequent students of administration have agreed with this conclusion, and the ideal span of control for a manager has typically been set at about six subordinates.

However, this rational conception of the individual manager's functions ignores certain problems which the narrow span of control creates. One of these is that in large organizations a narrow span increases the number of hierarchical levels required, thus accentuating problems of communication.[12] Research by Worthy indicates further disadvantages connected with a narrow span of control.[13] In the organization he studied, the span of control had been extended in deliberate violation of the old principle, and managers had several times the number of subordinates that had earlier been considered ideal. In addition to reducing the number of hierarchical levels in the organization, this innovation meant that managers could not depend too closely on their own superiors, and independence from superiors, as we have seen, fosters effective management practices.[14] At the same time, managers had too many subordinates to exercise close supervision over them, and this condition too was functional for operations, because close supervision (as we have noted often) impedes performance. Further, the broad span of control prevented the manager from becoming too closely involved with his subordinates, and such supervisory detachment, it will be recalled, tends to promote the loyalty of subordinates.[15] In short, comparing the significance of acute and obtuse pyramids of control, it appears that the latter exert several constraints on managers that are functional for effective supervision. The increased span of control prevents managers from engaging in certain practices—close supervision, dependence on superiors, involvement with subordinates—which may be tempting because of their short-range advantages but which prove to be detrimental to managerial effectiveness in the long run.

To measure the shape of the hierarchical pyramid, a simple index of the average span of control can be derived by the formula $S = \sqrt[L]{N}$,

[12] See Herbert A. Simon, *Administrative Behavior* (2d ed.), New York: Macmillan, 1957, pp. 26–28.

[13] Worthy, *op. cit.*, pp. 169–179.

[14] For another case where a broad span of control resulted in greater independence of middle managers, see Burton R. Clark, *Adult Education in Transition,* Berkeley: University of California Press, 1958, p. 72.

[15] An earlier study found that the smaller the work group, the less detached was the supervisor in evaluating his subordinates. Supervisors of small units tended to give higher and more uniform ratings to their subordinates than those of larger units. See Peter M. Blau, *The Dynamics of Bureaucracy,* Chicago: University of Chicago Press, 1955, p. 167.

where S is the span of control, L is the number of supervisory levels (the number of organizational levels above the operating level), and N is the total number of employees or members of the organization. Applying this formula to City Agency, which had 5 supervisory levels and about 1,000 employees, the span of control index is 4.0 (the fifth root of 1,000). County Agency had three supervisory levels and about 160 employees, yielding a span-of-control index of 5.4 (the cube root of 160). The index shows that the span of control for supervisors in County Agency was greater than that for City Agency supervisors. This difference is reflected on the lowest levels of the organizations in the size of the work groups, which contained five to six members in City Agency, but an average of seven caseworkers in County Agency.

Although this span-of-control measure is admittedly crude, it is useful as a reference against which variations between organizations as well as within organizations can be examined. For example, in the two welfare agencies the top managers had fewer subordinates than the number expected on the basis of the index, while the first-line supervisors had more than the expected number. This difference was probably due to the fact that top managers had many nonsupervisory responsibilities, such as representing the organization in its dealings with the environment, whereas first-line supervisors were chiefly engaged in overseeing the work of subordinates. In some organizations, however, the span of control of the higher officials is wider than that of the lower. In a plant studied by Dalton, for example, the number of superintendents under the assistant plant manager was greater than the number of subordinates under each of these superintendents.[16] This disparity reflects a division of labor in top management, with the second in command assuming primary responsibility for the internal operations of the plant, thus permitting the senior executive to devote most of his time and energy to external problems. The obtuse shape of the pyramid in the upper region of the hierarchy may well have been an important reason why middle management in this plant had much independence and maneuverability.

The range of managerial authority differs not only in terms of span of control but also in a different dimension, namely, the official's time perspective. Higher managers are generally expected to take a more long-range view than lower ones, taking into account not only the immediate but also the long-run consequences of decisions. This difference in perspective is related to differences in the way time is structured for the various levels in the organization. In the welfare agencies, for instance, the time of the caseworkers was highly fragmented. Their work was continually being interrupted, partly because

[16] Melville Dalton, *Men Who Manage,* New York: Wiley, 1959, chart facing p. 10.

emergency requests from clients urgently required action, but also partly as the result of supervisory practice. Supervisors felt free to disrupt the work of their subordinates by calling them to a conference or by telling them to stop what they were doing in order to start something else. In contrast, higher managers did not feel free to disrupt the work of their subordinates; accordingly, their conferences were scheduled by advance appointments. Neither do supervisors of more professionalized employees feel free to interrupt the work of their subordinates.[17] These differences suggest that the higher the level of responsibility—managerial or professional—the more control an employee tends to have over his own time.

Jaques has attempted to systematize these notions about the relationship between the freedom to decide how to use one's own time and the level of responsibility.[18] He recommends as a quantitative measure of responsibility the amount of time that elapses between successive reviews of an employee's performance by his superior. The time span during which a member of an organization makes decisions at his own discretion without accounting to his superior becomes, then, the operational definition of his level of responsibility.[19] Thus, a production worker may have his work checked by his foreman hourly, whereas the president of the company reports to the board of directors only annually. Jaques claims that his time measure of responsibility predicts salaries, and thus implies that management takes this factor implicitly into account when distributing financial rewards. This indicator also calls attention to the fact that exercising much responsibility involves the capacity to stand up under the uncertainty of not knowing the outcome or wisdom of one's own decisions for a period of time.[20] If the anxiety resulting from this uncertainty is so great that it impedes performance, it shows that an official has been given too much responsibility. Conversely, if an official finds his work routine and becomes impatient with his lack of discretion, it would indicate that he is ready for more responsibility than he has.

The physical distance separating the members of an organization is also relevant in this context. The frequency with which a superior checks on the performance of his subordinates depends not only on the amount of responsibility he or top management has explicitly

[17] See, for example, the discussion in Blau, *op. cit.*, p. 102. Supervisors of workers in a state employment agency felt free to interrupt the work of their subordinates, but supervisors of the more highly skilled federal investigators did not engage in such practices.

[18] Elliott Jaques, *Measurement of Responsibility,* London: Tavistock Publications, 1956, especially pp. 32–42.

[19] Alternatively, the longer the time during which the decisions of an official commit the resources of the organization, the higher his responsibility. See *ibid.*, pp. 23, 138.

[20] *Ibid.*, pp. 91–93.

decided to delegate to them but also on the ecological conditions, so to speak, that determine the opportunity for checking. On the one hand, subordinates may perform their duties continuously under the eyes of the superior; on the other hand, their work may carry them far from supervisory surveillance. If physical separation limits supervisors to intermittent contacts with subordinates, the latter often come to exercise more discretion than the official definition of their responsibility would warrant. Hughes, for example, calls attention to the increased level of unofficial responsibility that accrues to hospital nurses because their functional superiors, the physicians, are present for only short periods during the day and consequently exercise merely episodic control.[21] Clark reports that the geographical dispersion of the various centers of adult education in a metropolitan area resulted in a greater degree of autonomy for the officials in each of these centers.[22] Physical distance, then, decreases the degree of managerial control over subordinate units and hence increases the latter's autonomy. For it reduces the opportunities for checking on the performance of subordinates, just as a wide span of control does, and it is the actual frequency of such supervisory reviews rather than the official definition of the level of responsibility that governs the amount of discretion subordinates can and, indeed, must exercise.

From the standpoint of the management of an organization whose units are located in remote areas and dispersed over an entire country, these implications of physical distance constitute a challenge to devise long-range control mechanisms that maintain centralized control over local units and even over isolated members. In a recent study, Kaufman reports how the United States Forest Service has coped with this problem.[23] In addition to its elaborate manuals and guides, the Forest Service utilizes special techniques for detecting and discouraging deviations from official procedures among its scattered posts in the field. Rangers are asked to submit many written reports and to keep an official diary showing to the nearest half-hour how each work day is spent. They submit many statistics to be processed by regional headquarters, and while this processing is a service performed by headquarters for them, it also serves as a control device on ranger operations. Field inspections are regularly conducted covering both over-all and specialized technical programs. The public has the right to appeal decisions by rangers to the regional office. There are frequent transfers of Forest Service personnel from one post to another, a practice that reduces loyalty to a specific community and increases

21 Everett C. Hughes, *Men and Their Work,* Glencoe, Ill.: Free Press, 1958, p. 74.

22 Clark, *op. cit.,* pp. 71–72.

23 Herbert Kaufman, *The Forest Ranger,* Baltimore: John Hopkins Press, 1960.

dependence on the Service. These external controls over rangers are supplemented by internal constraints developed during the early years of training and fostered by personnel policies. Promotions are made only from within and are relatively slow, so that a man has acquired considerable experience and has become properly socialized before he is given much responsibility. Finally, the use of uniforms and badges and the relatively high status of the Service promote *esprit de corps* among the men and their identification with the organization. Although the Forest Service seems to have achieved centralized control over widely dispersed and isolated local units, the elaborate arrangements instituted for this purpose indicate that such control is not easily accomplished.

Staff and Line. The distinction between staff and line is one that has long informed the study of formal organizations. Line organization places emphasis on differences in rank, and its members have authority over production processes. Staff organization directs attention to specialization, and its members usually function in a research and advisory capacity.[24] In short, line officials possess formal authority, whereas staff members furnish specialized and technical advice to the appropriate line officials in the organizational hierarchy. On the basis of their job content and the functions they perform for unions, Wilensky has developed a typology of union staff experts which illustrates the variety of staff roles even in such relatively unspecialized organizations.[25] Types include the "Facts and Figures Man," who supplies leaders with technical, economic, and legal information, the "Contact Man," who furnishes intelligence on the political conditions and ideological tendencies in the society at large, and the "Internal Communications Specialist," who furnishes intelligence on the members of the organization and its operations that is needed for successful internal control.

At first glance it would appear that the division of labor between staff and line officials resolves two dilemmas of hierarchy discussed earlier: the blocking of communication due to hierarchical authority and the dilemma of disciplined compliance versus expert judgment. The staff, standing outside the hierarchical line, provides a mechanism for upward communication. Besides, staff personnel supply expert judgments which the line managers can transmit with confidence as authoritative directives, so that the decisions of the line can be governed by discipline, and those of the staff by expertness. These neat distinctions, however, are oversimplified and distort the actual situa-

[24] See Wilbert E. Moore, *Industrial Relations and the Social Order,* New York: Macmillan, 1947, pp. 96–104.

[25] Harold L. Wilensky, *Intellectuals in Labor Unions,* Glencoe, Ill.: Free Press, 1956, pp. 39–108.

tion. The staff expert can submit accurate operational reports to management, but if he presents adverse information concerning the actions of lower managers he will earn their antipathy and distrust, and these attitudes will prevent him from obtaining accurate information from them in the future. Alternatively, he can attempt to modify his report in order to maintain the good will of his informants, but in so doing he suppresses information that reflects adversely on lower managers, making his report hardly more accurate than their own communications.

The distinction between line authority and professional expertness is also less clear-cut in reality than the abstract concepts of line and staff make it appear. Increasingly, the line as well as the staff must make professional judgments. Thus, the engineer in charge of a production department in a factory is no less a technical expert than the accountant in charge of a staff department, and the latter is no less in a position of authority over a staff of subordinates than the former. As a matter of fact, in certain types of organizations, as Etzioni has noted, the roles of staff and line are reversed.[26] In the prototype of a formal organization, the line officers discharge the major functions of the organizations, and the professional staff, auxiliary functions. Accordingly, a factory may hire experts to do market research or public relations, but the main business of the organization —production—remains in the hands of line officials. Similarly, the line officers in an army are in charge of military action, and the staff merely furnishes assistance in making strategic and tactical decisions—intelligence about the enemy, supplies, and so forth. But in a university, the professional staff discharges the major institutional functions—teaching and research—while the "line" administrators look after the auxiliary functions, such as arranging schedules, housing students, and collecting fees. The same structure characterizes the hospital: the physicians who cure patients are on the professional staff and not line administrators.

Conflicts and Coalitions. Conflicts often develop between staff and line personnel. Dalton has discussed some of the sources of these recurring conflicts.[27] In part, conflicts between staff and line are due to differences in the background and career interests of the two types of personnel, but often they are not caused by a pursuit of self-interest or a clash of personal values but by the concern of officials with discharging their job responsibilities. For example, in City Agency, an accounting section was in charge of determining the exact amount of

26 Amitai Etzioni, "Authority Structure and Organizational Effectiveness," *Administrative Science Quarterly,* 4 (1959), pp. 43–67.

27 See Melville Dalton, "Conflicts between Staff and Line Managerial Officers," *American Sociological Review,* 15 (1950), pp. 342–351; and Dalton, *Men Who Manage, op. cit.,* pp. 73–108.

assistance due to a client. There were frequent conflicts between these staff auditors and the caseworkers, since the latter defined their responsibilities as providing service to clients (compared to auditors, even the procedure-oriented workers did so) whereas the auditors defined their responsibilities as protecting the taxpayers' funds and assuring that clients were not overpaid. Such conflicts are not motivated by self-interests; rather, they result from each party's devotion to its job and endeavors to discharge its responsibilities effectively.[28]

Dalton's early research emphasizes the conflicts between staff and line personnel, and his later work shows that conflicts also prevail among line officers (and presumably among staff officers).[29] In a study of four organizations—three factories and a department store—Dalton found that the most effective managers were goal-oriented rather than method-oriented.[30] The former were strong enough to fight for the effective accomplishment of their mission and for expanding the jurisdiction of their department. They were partly motivated by self-interest as they sought increased power and promotions, but theirs was not a self-interest that opposed organizational interests, since they strove to maximize departmental objectives. Seeking to expand the jurisdiction of one's department is a managerial responsibility, in government as well as in industry.[31] Since expansion tends to lead to border disputes, power struggles often developed between managers. In these battles the less aggressive managers lost out, and so did their departments.[32] Coalitions frequently developed in the course of such struggles, sometimes between managers on the same level, and sometimes between managers on different levels. Similar coalitions occur among army officers, as Janowitz has noted.[33]

Far from condemning these power struggles, Dalton looks on them as important mechanisms of organizational change. Formal organizations are frequently slow to adjust their official structure to changing conditions. Innovations that adapt operations to a new situation, and thus benefit the organization, are often made informally, sometimes

[28] Note that the same point was made in Chapter III from a somewhat different perspective and using different language. There we talked about internal conflicts being engendered by the fact that different departments in an organization are oriented to different publics. The auditors in the example above were apparently oriented primarily to serving the taxpaying public, while the caseworkers were concerned with providing service to their clients.

[29] Dalton, *Men Who Manage, op. cit.*, pp. 18–70.

[30] *Ibid.*, pp. 246–248. This distinction is somewhat similar to that between a professional and a bureaucratic orientation.

[31] On this point see Marshall E. Dimock, *The Executive in Action,* New York: Harper, 1945, pp. 53–68.

[32] Dalton, *Men Who Manage,* p. 34.

[33] Morris Janowitz, *The Professional Soldier,* Glencoe, Ill.: Free Press, 1960, pp. 293–294.

in violation of official procedures. Power struggles play a significant role in this process of adjustment to change in the organization, since they provide managers with incentives for making informal innovations which help them cope with operating problems and extend the scope of their influence.[34]

Dalton's image of organization managers, as the foregoing makes clear, contrasts sharply with Whyte's conception of the organization man.[35] Unlike Whyte, Dalton does not see the individual pressed by the organization into a mold, lacking initiative and avoiding conflicts. On the contrary, in Dalton's view the organization exerts pressure on managers to exercise initiative in the interest of discharging their responsibilities successfully and strengthening their position. Only those left behind in the struggle for positions of dominant influence become method-oriented organization men.

Crozier's study of plants in the French tobacco monopoly adds new insights to our discussion of power struggles between managers.[36] The plants studied were highly bureaucratized: almost every situation was governed by formal rules, jobs were allocated strictly along seniority lines, and candidates for positions were recruited from the outside through open competition. Interviewing in a number of different plants revealed that among the three categories of employees at the lower level of the firm the maintenance workers were very satisfied with their situation, production workers were less satisfied, and supervisors were quite dissatisfied. An identical pattern was found at the managerial levels of the organizations, where the technical engineers were both more satisfied and more aggressive than were the top managers, their superiors. There was considerable conflict between the engineers who were in charge of machine maintenance and other managers, and the former seemed to possess a good deal of informal power. In analyzing these patterns, Crozier suggests that bureaucratic formalization eliminates many areas of uncertainty. Power accrues to those who can control the remaining areas of uncertainty. In the bureaucratized tobacco plants, machine breakdown was the main area of uncertainty still remaining; hence, social processes expanded the unofficial power of maintenance workers and particularly of the engineers in charge of their activities. Their power position produced conflict

[34] Likert arrives at a similar conclusion. He considers conflicts between organization members essential and healthy as long as mechanisms are available to deal constructively with these conflicts, effective interaction is skillfully maintained by the personnel, and there exists a basic loyalty to the organization. See Rensis Likert, *op. cit.*, pp. 204–205.

[35] William H. Whyte, Jr., *The Organization Man,* New York: Simon and Schuster, 1956.

[36] Michael Crozier, *The French Bureaucratic System,* Stanford: Stanford University Press (forthcoming).

in the organization as others questioned the legitimacy of their controlling influence.

The major implication of Crozier's work seems to be that technical expertness is a source of power in formal organizations, just as Weber pointed out. But whereas Weber saw expertness as a basis of formal authority, Crozier views it as an alternative to it that becomes manifest in areas not yet fully bureaucratized. Moreover, the power that rests on expertness in areas of uncertainty is not stable, since "the expert's success is constantly self-diminishing."[37] As the knowledge of experts increases, it transforms areas of uncertainty into areas that can be routinely handled (for example, advances in medicine enable nurses to perform tasks that previously required the judgment of physicians). The expert's own success, therefore, reduces the base of his power or, at least, shifts its base to a new area of uncertainty.

Impersonal Mechanisms of Control

Assembly-Line Production. The concept of "impersonal authority" in formal organizations is usually interpreted to mean hierarchical relations free of personal involvement, in contrast, for example, to charismatic authority where personal ties bind the followers to their leader. Indeed, in our discussion of the supervisor's role we noted the significance of detachment for effective supervision. But impersonal authority may also mean management through nonhuman mechanisms of control. It is possible for management to design and install an impersonal control system in such a manner that it, rather than the hierarchy, exerts continuous constraints on the performance of subordinates. The assembly line exemplifies such a control mechanism.[38]

Once the production process on the assembly line has been designed, the conveyor exerts the major constraints on workers, indicating what job they are to do next, what materials and tools are to be used, and setting the pace at which the work must be done. The foreman in this situation is consequently relieved of many of his traditional responsibilities. In spite of this change, however, most foremen do not think that the assembly line has made their job easier, but stress that it creates new problems for them.[39] Faced with the moving conveyor and a certain output level to be maintained, worker turnover and absenteeism become crucial problems. The foreman must

[37] *Ibid.*

[38] See Charles R. Walker and Robert H. Guest, *The Man on the Assembly Line,* Cambridge, Mass.: Harvard University Press, 1952, pp. 9–14.

[39] See Charles R. Walker *et al., The Foreman on the Assembly Line,* Cambridge, Mass.: Harvard University Press, 1956, pp. 12–33.

quickly train new workers and must be able to obtain replacements speedily from his superior. While waiting for a replacement, he must redistribute the work load among the men present. Tools must be kept in good repair, supplies must be readily available if the work is to proceed on schedule, and the foreman must make the necessary arrangements, since the workers themselves are bound to their place on the line. Finally, although the conveyor assures that a certain amount of work will be done in a specified time, the supervisor must see to it that the work is of satisfactory quality. Note that all of these supervisory responsibilities except the last—maintaining performance quality—involve helping the workers rather than checking on them. The assembly line reduces the foreman's responsibilities for checking on workers and increases his responsibility for helping them.

Assembly-line production tends to reverse the flow of demand in the organization. Typically, demands flow downward in the hierarchy: orders are received from the level above and transmitted to the level below. And usually the foreman must direct operations by giving instructions and orders and by checking on work. An intensive downward flow of demands constitutes close supervision, a practice which tends to impede effective performance, as we have seen. Since the presence of a conveyor assures coordination and a certain level of productivity, it obviates the need for supervisory direction to achieve these ends. As a result, the major task of the foreman is transformed from directing and checking on subordinates to helping them and being their trouble-shooter. With the assembly line to organize the work process, the typical pattern of interaction is that the worker requests the foreman's help (whether he needs advice or merely supplies) rather than that the foreman makes demands on the worker—to do this job now or to work faster. Thus, the worker, not the foreman, initiates most of their interaction, reversing the flow of demand typical of close supervision in formal organizations. The same sort of reversal occurs on higher levels. The foreman, seeking services and materials, initiates most of the interaction with his superiors and staff officials. And in contrast to the frequent conflicts that occur between staff and line in other types of organizations, assembly-line foremen tend to look on the staff as being helpful.[40]

These and related changes cast the foreman on the assembly line in the role of adviser and assistant to his subordinates, and his new role affects his behavior even when he does make demands on workers. As noted, the foreman must still check on the quality of work performed, but his role as trouble-shooter is likely to induce him to seek improvements by offering advice to his workers rather than by issuing commands. For to issue curt orders would interfere with his

[40] *Ibid.*, pp. 34–36.

ability to maintain the role of trouble-shooter and helper, and hence impede his effectiveness as a foreman on the assembly line. In short, the assembly line, by removing some of the sources of conflict between foreman and worker, improves the relations between them.[41]

Performance Records. The evaluation of performance on the basis of statistical records also acts as an impersonal mechanism of control, as was observed in the study of a state employment agency.[42] The agency developed performance records as an indirect means of control—to provide the supervisor with information which would serve as a basis for the evaluation and guidance of workers. But once introduced, the records themselves became a direct mechanism of control. Since the records were public, they provided each employee with precise knowledge of how his performance compared with that of others and of his chances for getting a high performance rating. If his record was below average, this knowledge frequently motivated him to improve his performance. In this way public records which made comparative performance visible to officials controlled operations without any direct intervention on the part of the supervisor.[43] Such performance records can be used to control some qualitative as well as quantitative aspects of performance. Various types of errors and successes can be recorded in addition to the total number of cases processed. With some ingenuity, the quality of performance on quite complex tasks can be assessed by management. For example, to measure how carefully interviewers selected qualified applicants for jobs in the employment agency, the proportion of referrals who actually obtained the job to which they had been referred was computed. By including some statistical measures in the performance record and not others, and by placing more or less emphasis on any one measure, management could regulate the amount of effort interviewers would devote to various aspects of the work.

Observation in the employment agency revealed that these per-

[41] Although worker-foreman relations were good on the assembly line studied, this fact should not be taken to mean that workers were satisfied with their jobs. Most reported less satisfaction with present jobs than with previous jobs (not on the assembly line); the quitting rate for assembly plants in the automobile industry was higher than for other manufacturing industries; and there was some evidence that work on the assembly line generated aggression against the company. *Ibid.*, pp. 63, 120–121, 135.

[42] See Blau, *op. cit.*, pp. 33–48, and "Formal Organization," *American Journal of Sociology*, 63 (1957), pp. 62–63.

[43] It should, however, be noted that sometimes effective operations were impeded because making a good record became an end in itself. This situation was observed in the two welfare agencies as well as in the employment agency. In County Agency, for example, caseworkers sometimes worked on "easy" cases rather than on those with more urgent problems in order to increase the number of closings or to reduce the number of pending cases on their monthly statistical record.

formance records also reversed the flow of demand in the organization. The records provided supervisors with information about the performance of their subordinates and at the same time exerted direct influence on performance. In these circumstances the supervisor no longer needed to check frequently on the work of subordinates, and his task of critically evaluating their performance became easier and less onerous. He did not have to court the danger of antagonizing some of his subordinates by telling them that their work was inadequate, since he could, as one put it, "let the records speak for themselves." The information provided by performance records permitted the superior to supervise less closely and to let subordinates come to him with their problems. This reversed flow of demand was reflected in the finding that the more supervisors relied on statistical records, the less frequently did they initiate the social interactions between themselves and subordinates.

Even when the supervisor talked to an interviewer with a poor record about improving his performance, the existence of statistical records completely changed the nature of the conference. In the absence of performance records, the supervisor would have to start the conference by first telling the interviewer that his performance was inadequate. The existence of public records made this opening superfluous and, consequently, transformed the supervisor's discussion from a criticism that was likely to be resented to an offer to help the subordinate improve his record. Still another function of statistical records was that they served as an instrument for evaluating the work of interviewers on the basis of accomplished results, and such an evaluation system minimized the significance of detailed operating procedures on how to accomplish results and thus enabled management to permit officials more discretion in their work. The freedom from close supervision and from rigid operating rules that performance records make possible tends to enhance work satisfaction.

We have briefly discussed some of the consequences of two mechanisms of impersonal control: the assembly line and statistical records of performance. Note that these two mechanisms differ greatly in form and were observed in two very different kinds of organizations. Yet since both served as mechanisms for exerting impersonal constraint, their consequences in the organizations were quite similar. Both were in part substitutes for control through personal supervision; both reversed the flow of demand in the organization; and both improved relations between supervisors and subordinates. However, there was an important difference. Statistical records involved the evaluation of performance on the basis of results achieved, whereas the assembly line did not, and only the statistical performance records increased the discretion of employees and hence their job satisfaction. The

assembly line had just the opposite consequences. It reduced the discretion of operators to a minimum, and as a result job satisfaction was extremely low.

Automation. Increased mechanization and specialization, exemplified by assembly-line production, has had the effect of reducing the skill level of workers as well as their discretion.[44] Skilled craftsmen have been replaced by semiskilled machine operators. However, further mechanization of the factory, particularly automation, has been expected to raise the level of skill and responsibility of operators again, since it would require that the modal worker be an expert mechanic who can maintain the complex machines. Specifically, we expected automation to serve as an impersonal mechanism of control more akin to performance records than to the assembly line. The same functions as those served by the assembly line—effecting coordination and maintaining rate of production with no need for close supervision—were expected from automation, but without the same dysfunctions—lack of discretion and low job satisfaction—because the worker would be highly skilled and could be evaluated on the basis of results, for example, his record of machine maintenance.

Studies of automated factories do not bear out these expectations. On the basis of interviews conducted with a random sample of workers in an automated automobile factory, Faunce discovered that workers felt they were more closely supervised than they had been in regular assembly-line automobile plants.[45] Workers reported more contacts with foremen,[46] and they were less satisfied with their relations with foremen. In addition, the number of foremen per worker had been increased in consequence of the greater complexity of the machines, their greater interdependence, and especially the costly delays occasioned by their breakdown. Also contrary to expectations is the finding that automation restricted the social contacts between workers even more than had the assembly line. In contrast to the assembly line, there were distinct social groups, but even so there was less chance for social interaction among workers in the automated plant. Faunce suggests that reduced interaction was due to the closer attention required by the work, the greater distances between workers, and the greater noise of the machines.

[44] For an early study of the breakdown of the skill hierarchy and its consequences for worker discretion and satisfaction, see W. Lloyd Warner and J. O. Low, *The Social System of the Modern Factory*, New Haven: Yale University Press, 1947, pp. 66–89.

[45] William A. Faunce, "Automation in the Automobile Industry," *American Sociological Review*, 23 (1958), pp. 401–407.

[46] No information was presented on who initiated these contacts, so that we do not know how or whether the flow of demand was affected.

A study by Simpson of a mechanized spinning department in a textile mill provides an interesting contrast with the foregoing case of automation.[47] Simpson found no evidence of close supervision in the spinning department, noting that first-line foremen had relatively few work-related contacts with their subordinates but interacted primarily with other foremen on their own level. This observation parallels the conclusion we deduced from the study of assembly-line foremen—that mechanized techniques which pace the work make close supervision unnecessary—but it differs from the situation observed in the automated factory, where supervision was quite close. Simpson interprets these results by suggesting that some degree of mechanization, such as that characteristic of the spinning department and of the assembly line, reduces the need for close supervision because the production process exerts some of the constraints on worker behavior that otherwise would have to be exerted by foremen. However, further mechanization, as occurs in automation, again increases the need for supervision and hierarchical communication because in the interest of effective operations the recurrent and very costly breakdowns of the complex machinery must be minimized and quickly dealt with when they do occur. The implication is that the operators in an automated plant cannot assume responsibility for machine maintenance by making minor repairs themselves and calling in expert mechanics when necessary.

Walker's study of a semiautomatic steel mill furnishes some explanation for this interpretation as well as confirmation of many of Faunce's findings.[48] Since workers had been transferred from another steel mill, it was possible to compare their new social situation and experiences with their former ones in a mechanized but nonautomatic plant. Workers reported greater social isolation in the new mill than in the old, and most felt that they were physically more isolated in the new plant. At first their attitudes toward supervision improved, but soon they deteriorated sharply. After a period of adjustment, attitudes to supervisors again improved but remained negative. No information was obtained on the closeness of supervision. In general, attitudes were also negative toward management. Workers felt that no attempt was made to improve undesirable physical conditions, such as excess smoke, and that they were not getting a fair share of the profits realized by the improved technology; they also feared that automation might result in layoffs during a depression.[49] With regard to

[47] Richard L. Simpson, "Vertical and Horizontal Communication in Formal Organizations," *Administrative Science Quarterly*, 4 (1959), pp. 188–196.

[48] Charles R. Walker, *Toward the Automatic Factory*, New Haven: Yale University Press, 1957, pp. 47, 51, 93.

[49] *Ibid.*, pp. 81, 94–96, 184–185.

their work, the men said they experienced less physical fatigue but more mental strain owing to tensions created by watching fast-moving machines closely. Their reported work satisfaction was only fair, but it was probably higher than that connected with assembly-line work. Surprisingly, the average skill level of the workers had not increased. There were fewer high job classifications in the automated plant than in the old one, and there was less opportunity for advancement, which was a source of great dissatisfaction to the workers.[50] The low skill level of operators of automated machines—not superior to that of other factory workers—helps to explain why responsibility for machine maintenance could not be assumed by these workers themselves but only by foremen.

Finally, a case history of the introduction of electronic data-processing equipment into the accounting division of a large light and power company is of interest because it provides some information about the effects of automation on white-collar groups.[51] As in the factory studied by Walker, the introduction of automation into the white-collar office did not significantly increase the average job grade or skill level. While most routine and menial jobs were eliminated by the change-over, so were a number of high-level non-supervisory jobs. The lack of improvement in job grades was a source of dissatisfaction among both workers and first-line supervisors. The new equipment also reduced discretion, particularly in the middle levels of the organization, and allowed less autonomy in setting the work pace for both groups and individuals. The greater interdependence of operations created problems of coordination and led to greater centralization of decision-making in the upper levels of the organization.[52]

A conclusion that emerges from these studies on the social consequences of automation is that as long as automation is not accompanied by a higher level of skill on the part of the workers involved, it does not result in increased discretion for them; hence, job satisfaction is not notably improved by such automation and may in fact be reduced. Given a low level of skill of operators, moreover, the complexity of the machines and the interdependence of operations appear to require closer supervision and greater centralization of decision-making to minimize the costly delays occasioned by machine breakdowns or by imperfect coordination. It may be that further automation and, particularly, a reorganization of the work force to assure that

[50] *Ibid.*, pp. 29–39, 59–61, 97–99, 192–193.

[51] Floyd C. Mann and Lawrence K. Williams, "Observations on the Dynamics of a Change to Electronic Data-Processing Equipment," *Administrative Science Quarterly,* 5 (1960), pp. 217–256. See also Floyd C. Mann and L. Richard Hoffman, *Automation and the Worker,* New York: Holt, 1960.

[52] Mann and Williams, *op. cit.,* pp. 247–252.